

Shiye (Sally) Cao

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RESEARCH OVERVIEW

My goal is to enable robots to converse effectively, naturally, and robustly with people by advancing three aspects of conversation: what robots say, how they deliver it, and how they repair breakdowns. My long-term vision is for these conversational capabilities to enable robots to provide sustained support in people's everyday lives.

Keywords: human-computer interaction, embodied AI, privacy and security.

EDUCATION

Johns Hopkins University, Baltimore, Maryland, USA

PhD Candidate in Computer Science advised by [Chien-Ming Huang](#) and [Anqi Liu](#) 2022–Present

M.S.E in Computer Science 2021–2022

B.S. in Computer Science and Applied Mathematics and Statistics 2018–2021

PUBLICATIONS

*Indicates authors contribute equally to the work

All publications can be found on my [Google Scholar](#) profile. Top-tier conferences (HRI, RSS, CHI, CSCW) have acceptance rates around 25% and are considered premier venues for Computer Science research.

Peer-Reviewed Conference Papers

C.7 **ERR@ HRI 3.0 Challenge: Multimodal Detection of Errors and Anticipation in Human-Robot Interactions**

Maria Teresa Parreira, Micol Spitale, Maia Stiber, **Shiye Cao**, Amama Mahmood, Chien-Ming Huang, Hatice Gunes, Wendy Ju

Proceedings of the 28th International Conference on Multimodal Interaction (ICMI'26)

<https://arxiv.org/abs/2607.11570>

C.6 **ELLA: Generative AI-Powered Social Robots for Early Language Development at Home**

Victor Nikhil Antony*, **Shiye Cao***, Shuning Wang, and Chien-Ming Huang

Proceedings of the 25th Annual ACM Interaction Design and Children Conference (IDC'26)

<https://dl.acm.org/doi/10.1145/3773077.3806113>

C.5 **Reframing Conversational Design in HRI: Deliberate Design with AI Scaffolds**

Shiye Cao, Jiwon Moon, Yifan Xu, Anqi Liu, and Chien-Ming Huang

Proceedings of 2026 ACM/IEEE International Conference on Human-Robot Interaction (HRI'26) | Acceptance rate: 23.2%

<https://dl.acm.org/doi/10.1145/3757279.3785640>

C.4 **ERR@ HRI 2.0 Challenge: Multimodal Detection of Errors and Failures in Human-Robot Conversations**

Shiye Cao, Maia Stiber, Amama Mahmood, Maria Teresa Parreira, Wendy Ju, Micol Spitale, Hatice Gunes, and Chien-Ming Huang

Proceedings of 33rd ACM International Conference on Multimedia (ACM-MM'25)

<https://dl.acm.org/doi/abs/10.1145/3746027.3762073>

C.3 **Interruption Handling for Conversational Robots**

Shiye Cao*, Jiwon Moon*, Amama Mahmood, Victor Nikhil Antony, Ziang Xiao, Anqi Liu, and Chien-Ming Huang

Proceedings of the 2025 Robotics: Science and Systems Conference (RSS'25)

<https://www.roboticsproceedings.org/rss21/p089.pdf> | Acceptance rate: 27.4%

C.2 **Voice Assistants for Health Self-Management: Designing for and with Older Adults**

Amama Mahmood, **Shiye Cao**, Maia Stiber, Victor Nikhil Antony, and Chien-Ming Huang

Proceedings of 2025 ACM Conference on Human Factors in Computing Systems (CHI'25)

<https://dl.acm.org/doi/10.1145/3706598.3713839> | Acceptance rate: 25.1%

C.1 **“What If It Is Wrong”: Effects of Power Dynamics and Trust Repair Strategy on Trust and Compliance in HRI**

Ulas Berk Karli*, **Shiye Cao***, and Chien-Ming Huang

Proceedings of 2023 ACM/IEEE International Conference on Human-Robot Interaction (HRI'23)

<https://dl.acm.org/doi/10.1145/3568162.3576964> | Acceptance rate: 25.3%

Peer-Reviewed Journal Articles

J.4 **Designing for Appropriate Reliance: The Roles of AI Uncertainty Presentation, Initial User Decision, and User Demographics in AI-Assisted Decision-Making**

Shiye Cao, Anqi Liu, and Chien-Ming Huang

Proceedings of the ACM on Human-Computer Interaction (CSCW'24)

Volume 8, Issue CSCW1, Article 41, Pages 1-32

<https://dl.acm.org/doi/10.1145/3637318>

J.3 **How Time Pressure from Different Phases of Decision-Making Influences Human-AI Collaboration**

Shiye Cao*, Catalina Gomez*, and Chien-Ming Huang

Proceedings of the ACM on Human-Computer Interaction (CSCW'23)

Volume 7, Issue CSCW2, Article 277, Pages 1-26

<https://dl.acm.org/doi/10.1145/3610068>

J.2 **Crowdsourcing Thumbnail Captions: Data Collection and Validation**

Carlos Aguirre*, **Shiye Cao***, Amama Mahmood, and Chien-Ming Huang

ACM Transactions on Interactive Intelligent Systems (TiiS, 2023)

Volume 13, Issue 3, Article 14, Pages 1-28

<https://dl.acm.org/doi/10.1145/3589346>

J.1 **Understanding User Reliance on AI in Assisted Decision-Making**

Shiye Cao and Chien-Ming Huang

Proceedings of the ACM on Human-Computer Interaction (CSCW'22)

Volume 6, Issue CSCW2, Article 471, Pages 1-23

<https://dl.acm.org/doi/10.1145/3555572>

Refereed Symposium & Workshop Papers

W.4 **Rapid Conversational Authoring of Interactive Human-Robot Activities**

Shiye Cao*, Aureliane Tchamdja*, Chien-Ming Huang

To appear at the AAAI Fall Symposium on Unifying Representations for Robot Application Development (UR-RAD'26)

W.3 Long-Term In-Home Robotic Support for Children with ASD

Shiye Cao

2026 HRI Pioneer Workshop | Acceptance rate: 23%

<https://dl.acm.org/doi/10.1145/3776734.3794629>

W.2 Practical Trade-offs in Generative Robot Expression Design

Victor Nikhil Antony, **Shiye Cao**, Chien-Ming Huang

Expressive robot faces with Vizij: a workshop / tutorial hybrid at HRI 2026

<https://drive.google.com/file/d/1oZPRrEJ829Ju4gAeAPaSNcZOqEjEAp9b/view>

W.1 Eyes Are the Windows to AI Reliance: Towards Real-Time Human-AI Reliance Assessment

Shiye Cao, Shichang Ke*, Alexandra Mo*, Anqi Liu, and Chien-Ming Huang

2023 CHI Workshop on Trust and Reliance in AI-Assisted Tasks (TRAIT'23)

https://chi-trait.github.io/papers/2023/CHI_TRAIT_2023_Paper_35.pdf

Under-Review Papers

U.3 Privacy and Security Communication for Robots: Current Practice, Open Challenges, and Recommendations

Shiye Cao, Laura Fleig, Yaxing Yao, Chien-Ming Huang

U.2 Towards Child-Centered Error Management In-the-Wild: Children's Reactions and Recovery Following Robot Errors

Shiye Cao*, Shuning Wang*, Victor Nikhil Antony, Leyun Kim, Chien-Ming Huang

U.1 Sync or Sink: Conversational Pace Matching for Human-Robot Collaboration

Yifan Yin*, Binze Li*, **Shiye Cao***, Katherine Guerrierio, Seokhyun Baek, Tianmin Shu, and Chien-Ming Huang

AWARDS

Best Poster Award, Third Annual Celebrating Women in Data Science and AI Symposium

[2026](#)

HRI Pioneer

[2026](#)

CRA Outstanding Undergraduate Researcher Finalist

2022

TEACHING EXPERIENCES

Teaching Assistant, Introduction to Human-Robot Interaction

Spring 2026

Teaching Assistant, Introduction to Human-Computer Interaction

Fall 2022

Course Assistant, Machine Learning

Fall 2020, Spring 2021, Fall 2021

PROFESSIONAL SERVICE

Organizing Committee: ICMI'26 ERR@HRI 3.0 Challenge, ACM-MM'25 ERR@HRI 2.0 Challenge

Program Committee: 2027 HRI Pioneers Workshop, NeurIPS '22 Gaze Meets ML Workshop

Reviewer: HRI'26, CHI'26 (Outstanding Review), CSCW'25, CHI'25, HRI'25, HRI'24 (Outstanding Review), CSCW'24, CSCW'23, THRI

Student Volunteer: HRI'23, CSCW'23, CHI'23, CSCW'22

Community Outreach:

- Organized outreach event at the Applied ABC Autism Therapy Center (4/2026)
- Organized outreach event at the Port Discovery Children's Museum (6/2026)
- Organized outreach event at the Henderson Hopkins School (8/2026)

RESEARCH MENTORING

I have mentored nine students on projects leading to publications. Mentees have continued to PhD programs at Yale and the University of Chicago, and one received a CRA Outstanding Undergraduate Researcher Honorable Mention.

PhD Students

Yifan Yin (JHU CS PhD)	2026
Publications: U.1	
Binze Li (JHU CS PhD)	2026
Publications: U.1	
Laura Fleig (JHU CS PhD)	2026
Publication: U.3	

Master's Students

Ulas Berk Karli (JHU Robotics MSE)	2021–2023
Next: CS PhD at Yale University	
Publications: C.1	
Yifan Xu (JHU CS MSE)	2025–2026
Publications: C.5	
Shuning Wang (JHU Systems Engineering MSE)	2025–2026
Publications: C.6, U.2	

Undergraduate Students

Jiwon Moon (JHU CS BS)	2023–2025
Next: CS PhD at the University of Chicago	
 2025 CRA Outstanding Undergraduate Researcher Honorable Mention	
Publications: C.3, C.5	
Leyun Kim (JHU CS & Cognitive Science BS)	2026
Publications: U.2	
Aureliane Tchamdja (UMBC CS BS, NSF REU)	2026
Publications: W.4	

Selected Press

[Celebrating women in data science and AI](#) (JHU CS News)

[Aging gracefully with AI](#) (JHU CS News)

[Talking robots learn to manage human interruptions](#) (JHU HUB)

[CS student named finalist for 2022 CRA Outstanding Undergraduate Researcher Award](#) (JHU CS News)